Incompressible Aerodynamics

References:

1. Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.
2. MAE 3302 *Incompressible Aerodynamics* (Fall 2001), by Professor Lund.

1. Why is the energy equation not needed in the study of incompressible flow?

   For constant density, there is no exchange between the mechanical (pressure potential and kinetic) energy and the internal (thermal) energy. As a result, the mass and momentum conservation equations are sufficient to solve for the velocity and pressure.

2. What is the Bernoulli equation? When is it valid?

   The Bernoulli equation says that the pressure plus the kinetic energy ($p + (1/2)\rho_\infty U^2 = \text{constant}$) is constant along a streamline. It is valid for steady, incompressible, inviscid flow.

3. What is the interpretation of the Bernoulli equation from an energy conservation standpoint?

   The sum of the pressure potential energy and kinetic energy remains fixed (The first law of thermodynamics for this simple system).

4. Which component of pressure is constant along a streamline in an incompressible, inviscid, steady flow? How do you know this?

   The stagnation pressure. From Bernoulli's equation: $p + (1/2)\rho_\infty U^2 = \text{constant}$

5. Nearly all aircraft use a pitot-static tube to measure airspeed. How does this device work? Will it still work if the flight Mach number is greater than 0.3? Explain.

   The airspeed is proportional to the square root of the difference between the stagnation and static pressures. The pitot tube can be used for M > 0:3, but the equation relating the airspeed to the pressure difference is more complicated.

6. How is the pressure coefficient related to the velocity for steady, incompressible, inviscid flow?

   Through the non-dimensional form of the Bernoulli equation:

   $$C_p = 1 - (U/U_\infty)^2$$

7. What value does the pressure coefficient take at a stagnation point for steady, incompressible, inviscid flow? What value does it take in the free stream?

   $C_p = 1$ at a stagnation point and $C_p = 0$ in the freestream.

8. When is the pressure coefficient negative?

The pressure coefficient is negative when the local velocity is greater than the free stream speed.

9. What condition must the velocity field satisfy if the flow is incompressible? Which conservation law does this condition come from?

The divergence of velocity is zero ($\nabla \bullet \vec{V} = 0$). This result comes from the mass conservation law.

10. The velocity potential will satisfy Laplace's equation provided that two conditions are met.

The flow is irrotational and incompressible.

11. What additional condition must be met if the stream function is to satisfy Laplace's equation also?

The flow must be two-dimensional.

12. What simplification in the solution to aerodynamics arises from the fact that Laplace's equation is linear?

Elementary solutions may be superimposed to yield more complex solutions.

13. What boundary condition must the velocity field satisfy at infinity?

The velocity must revert to the freestream condition.

14. What boundary condition must be satisfied by an inviscid fluid at a solid surface? Is this condition ever met in nature? Why doesn't the inconsistency of this boundary condition lead to serious problems in aerodynamics?

The velocity must be parallel to the surface. This condition is not met in nature since viscosity ensures that the flow is brought to rest at a solid surface. The inviscid boundary condition is an approximation where the body surface is translated to the edge of the boundary layer, where the flow tangency condition is valid. For high Reynolds numbers, this approximation is acceptable since the boundary layers are thin compared with the body thickness.

15. What is meant by an elementary flow solution? How can elementary solutions be used to solve problems in aerodynamics?

It is a solution to Laplace's equation that results in a relatively simple flow pattern. Elementary solutions are superimposed together to give more complex solutions.

16. What is the velocity potential and stream function for a uniform flow in the x-direction?

$$\psi = U_\infty y \text{ and } \phi = U_\infty x$$

17. Many of the elementary flow solutions are referred to as singularities. Why is this?

Since they become singular (infinite) at one or more points in the field.

18. Why doesn't the singular nature of the elementary flow solutions spoil the solution procedure?

    Because the singularities are almost always inside the closed streamline that mimics the body surface (i.e. not contained in the region of interest).

19. What potential flow assumption does the source violate? Where does the violation occur?

    Mass conservation is violated at r = 0.

20. What physical quantity is the constant in the expression for the source related to?

    The volume flow rate (mass flux divided by density).

21. A closed streamline is formed when a uniform flow, a source, and a sink are superimposed. How is this flow pattern related to the flow over a similarly shaped solid object immersed in the flow?

    The flow outside the closed streamline will be the same as the flow around the real solid object.

22. How is a doublet formed?

    By superimposing a source and sink in a limiting process, the strengths become infinite as the distance between them is reduced.

23. Does the streamline pattern of a doublet resemble the elementary flow solutions that are used to generate it? Explain.

    Not really. The source-sink combination results in a circulation pattern that is rather different from the purely radial flow of a source or sink.

24. How can elementary flow solutions be used to determine the flow over a circular cylinder?

    Superimpose a doublet and free stream.

25. What values of lift and drag coefficients are predicted by potential flow theory for flow around a circular cylinder? Is the analysis correct? What effect is missing?

    Lift and drag are both zero (assuming that the cylinder is not spinning). Lift is correct, drag should not be zero since viscosity will contribute to friction drag as well as cause flow separation that leads to pressure drag.

26. Will a more physically correct analysis of the cylinder flow change the predicted values of the drag coefficient? How about the lift coefficient?

    Yes, a viscous analysis will yield non-zero drag. The lift should remain zero.

27. What potential flow assumption does the vortex violate? Where does the violation occur?

    Irrotationality is violated at r = 0.

28. What elementary flow solution must be added to the circular cylinder flow in order to generate a lifting force? What physical situation does the resulting flow correspond to?

A vortex. This corresponds to a spinning cylinder in a viscous fluid.

29. Why are spinning cylinders typically not used as lifting devices in aeronautical applications?

Since considerable power is required to spin the cylinder at the required speed.

30. Why is the panel method preferred for bodies of arbitrary shape?

Since it is a relatively simple means of generating the flow around any body shape. The solution can be made as accurate as desired by increasing the number of panels.

31. How can the source panel method be used to compute the pressure distribution on an aerodynamic body of the prescribed shape?

Cover the surface with source panels, determine their strength such that the flow tangency boundary condition is satisfied, determine the surface velocity from the resulting potential flow solution, and determine the pressure from Bernoulli's equation.

32. What is a source sheet? How is it used in the source panel method?

It is a two-dimensional surface over which a continuous distribution of sources is placed. It forms the basis of the source panel method where each panel is actually a source sheet.

33. How can one determine the net velocity potential generated by a source sheet distributed over the surface of an aerodynamic body?

Integrate the contributions to the potential due to all points on the source sheet.

34. What is a source panel? How is the panel approximation used to simplify the task of finding the correct source distribution for the given body shape?

It is a source sheet where the strength is either constant or simply varying (linear, say). The panel approximation allows the job of finding the source distribution to be reduced to solving a linear system of equations for the panel strengths.

35. What boundary condition is used to obtain a system of equations for the source panel strengths?

Flow tangency on the surface of the body.

37. How are the velocity components determined at an arbitrary point in the field when the source panel method is used?

By summing up the velocities induced by each of the individual panels in the system and then adding the free stream if any.

38. How is the component of velocity normal to the surface computed in the source panel method?

The velocity components are found as in the previous question with the field point taken to be the center of the panel. The normal component of this velocity is then found by taking the dot product with the unit normal vector.

39. The quantity I_{ij} is the aerodynamic influence matrix. What does this matrix represent? What important role does it play in the source panel method?

It gives the velocity normal to the i^{th} panel induced by the j^{th} panel. It forms the heart of the panel method since the objective is to find the source strengths so that the normal component of velocity is zero at all control points.

40. The source panel method results in what type of equation to be solved on the computer? Do you think that it is difficult to program a computer to perform these operations?

A linear system of equations. The computer can solve this system with ease.

41. Will a solution via the source panel method violate mass conservation when the flow external to a closed body is considered? Explain your answer.

No. The solution will automatically satisfy the condition that the sum of the source strengths is zero.

$$\sum_{i=1}^{N} \lambda_j = 0$$

42. Potential flow theory predicts zero drag for a circular cylinder. Is this correct? What two sources of drag should really be present at moderate to high Reynolds numbers?

No. Friction and pressure drag are present in reality.

43. What causes the large drop in drag coefficient for a circular cylinder as the Reynolds number increases through about 3×10^5.

A transition from laminar to turbulent boundary layers on the cylinder surface. Turbulent boundary layers are more resistant to separation and thus the cylinder produces a smaller wake in this case.

44. What causes the poor agreement between the theoretical and measured pressure distributions for a circular cylinder?

Differences in the ideal and actual flow pattern. The real flow results in a separation and large wake behind, where viscous effects convert useful energy to heat.

45. What is D'Alembert's paradox? How is it reconciled?

Drag on an inviscid body is zero. One needs to include viscous effects to get correct predictions of drag.

46. State the Kutta-Joukowski theorem.

The lift on a body is equal to the free-stream density times the free-stream velocity times the circulation around the body (i.e. $L' = \rho U_\infty \Gamma$).

47. How can the Kutta-Joukowski theorem be used to simplify the task of computing the lift on an airfoil?

Only the circulation needs to be determined; it is not necessary to find and integrate the pressure distribution.

48. What quantity is discontinuous across the vortex sheet? How is the size of this jump related to the strength of the sheet?

The velocity tangent to the sheet. The jump in velocity is equal to the local value of the circulation density.

49. How is the vortex sheet used in airfoil theory? How is the lift ultimately determined in this approach?

Cover the airfoil surface with a vortex sheet. Determine its strength distribution so that the airfoil surface is a streamline. The lift is computed from the Kutta-Joukowski law using the net circulation.

50. What approximation is made in thin airfoil theory? How does this simplification simplify the analysis?

The airfoil is sufficiently thin that the effects of thickness can be ignored. This allows the problem to be reduced to finding the circulation distribution on the chord line.